systems

MDPI

Article

Measurement, Regional Disparities, and Spatial Convergence in the Symbiotic Level of China's Digital Innovation Ecosystem

Shengnan Li [1], Zhouzhou Lin [2,*], Yingwen Wu [3] and Yue Hu [2]

[1] School of Economics and Management, Nanjing Tech University, Nanjing 211816, China; lishengnan@njtech.edu.cn
[2] School of Politics and Public Administration, Soochow University, Suzhou 215123, China; 20224202038@stu.suda.edu.cn
[3] School of Future Science and Engineering, Soochow University, Suzhou 215222, China; ywwu@suda.edu.cn
* Correspondence: zzlin@suda.edu.cn

Abstract: Based on the panel data of 30 provinces in China from 2013 to 2022, this paper constructs a measurement index system for the symbiotic level of digital innovation ecosystems from three dimensions: the symbiosis of digital innovation subjects, the digital innovation environment, and digital innovation interaction. This paper applies the entropy weight TOPSIS method, Dagum Gini coefficient decomposition, and spatial convergence analysis to empirically examine the symbiotic levels, regional disparities, and spatial convergence of China's digital innovation ecosystem. The results are as follows: (i) At the national level, the symbiotic level of China's digital innovation ecosystem has generally increased, creating a spatial distribution pattern that is "high in the east, flat in the middle, and low in the west". (ii) From a regional perspective, the major disparities between regions are the primary factors contributing to the overall difference in the symbiotic level of China's digital innovation ecosystem. (iii) From the perspective of σ convergence, regional disparities in the symbiotic level of the digital innovation ecosystem are constantly expanding, and uneven regional development is intensifying. (iv) From the perspective of absolute β convergence, regions with lower levels of symbiosis in the digital innovation ecosystem have a faster growth rate of symbiosis than regions with higher levels of symbiosis, and there is a certain spatial spillover effect. (v) From the perspective of conditional β convergence, economic structure and innovation application can accelerate the spatial convergence of China's digital innovation ecosystem symbiosis to a certain extent.

Keywords: digital innovation ecosystem; symbiotic level; regional disparities; spatial convergence

check for updates

Academic Editor: Mitsuru Kodama

Received: 13 February 2025
Revised: 19 March 2025
Accepted: 28 March 2025
Published: 4 April 2025

Citation: Li, S.; Lin, Z.; Wu, Y.; Hu, Y. Measurement, Regional Disparities, and Spatial Convergence in the Symbiotic Level of China's Digital Innovation Ecosystem. *Systems* **2025**, *13*, 254. https://doi.org/10.3390/systems13040254

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

As the basic component of the national innovation system, the innovation ecosystem effectively integrates the two strategic ideas of "ecological priority" and "green development" and is an important carrier of practical innovation-driven development [1,2]. Unlike traditional innovation systems, which focus mainly on the content of innovation elements, innovation ecosystems focus more on the structural symbiotic relationships between innovation subject elements and between innovation subject elements and innovation environment elements within the system [3–5]. With the advent of the digital economy, China's innovative development has entered the digital era [6]. Digital innovation has reshaped overall social productivity with the Internet of Everything and highly socialized

resource allocation, breaking the space–time limit in the process of traditional enterprise organization production and service and becoming a new energy source for the development of new quality productivity [7]. According to a report on the digital economic development of China's regional and urban areas, the scale of China's digital economy reached CNY 5.02 billion in 2022, which is equivalent to that of a secondary industry. However, the competitiveness index of the digital economy between regions shows uneven characteristics. This indicates that the large-scale expansion of the digital economy has been unable to meet the needs of high-quality development in China, and the construction of digital innovation ecosystems has become the key to the sustainable and balanced development of the digital economy at this stage [8]. The digital innovation ecosystem integrates elements such as digital technology, artificial intelligence, and big data on the basis of the original innovation ecosystem. It not only leverages the cross-spatial continuity of digital innovation and promotes the close connection of various digital innovation subjects within the system but also follows the laws of self-organizing evolution, accelerates the symbiosis of system elements [9,10], and better empowers the emergence of new quality productivity [11,12]. However, owing to the current unequal income distribution and fuzzy communication and cooperation, the overall development level of China's digital innovation ecosystem is relatively low [13,14]. Faced with the dual requirements of high-quality development and new quality productivity development, the issue of how to effectively improve the symbiotic level of China's digital innovation ecosystem has become a real dilemma to be solved at this stage.

In fact, it is highly necessary and scientific to study the regional digital innovation ecosystem from the perspective of symbiosis. On the one hand, regional digital innovation ecosystems are ecological, organic-style dynamic systems consisting of different symbiotic units that interact and are interdependent through the cross-spatial connectivity characteristics of digital innovation and are influenced by the symbiotic environment [15]. Promoting the symbiosis of regional digital innovation ecosystems can effectively enhance the close integration of science and technology with the economy, innovation, and business and realize the creation and value-added of scientific worth, which has become the preferred strategy for regional innovation development [16]. On the other hand, the symbiosis of a regional digital innovation ecosystem is a measure of the overall situation presented by the interaction, interdependence, and interactive evolution of the elements of digital innovation ecosystems from the perspective of ecology. It not only shows the interaction between symbiotic units but also includes the overall effects of resource exchange, energy flow, relationship evolution, and comprehensive interaction. Therefore, exploring ways to promote the symbiosis of regional digital innovation ecosystems has important scientific value for promoting the high-quality development of regional digital innovation ecosystems. Moreover, it is a key way to improve the level and quality of regional management [17].

Many scholars have studied digital innovation ecosystems. From the perspective of research topics, some studies have focused on the value cocreation of digital innovation ecosystems [18,19], niche suitability [20], network application program interfaces [21], etc. From the perspective of influencing factors, some studies have shown that the development of digital innovation ecosystems is affected by many factors, such as the digital space [22], stakeholder cooperation strategies [23], and watershed compensation policies [24]. From the perspective of research scope, Wolfert et al. (2023) took Europe as the research scope and discussed the design principles and organizational framework of agricultural product digital innovation ecosystems [25]. Randhawa et al. (2024) studied the legalization of digital technology in the open innovation ecosystem in Belgium [26], and da Rosa et al. (2024) took Brazil as the research scope and empirically investigated the single or combined impact of hotel digital capability and innovation ecosystem cooperation and competition on

digital innovation and reducing food waste [27]. Many previous studies have demonstrated the importance of digital innovation ecosystems for regional development. However, at the same time, some problems can be identified. First, for the regional digital innovation ecosystem, the symbiosis of its elements is the core issue. In particular, the symbiosis level of a regional digital innovation ecosystem directly affects its development quality; however, the current research on the symbiosis level of regional digital innovation ecosystems is insufficient. Second, as the world's second largest economy, China has a vast territory, which objectively leads to regional diversity and differences, such as the level of economic development, geographical location, and policy background. A systematic study of the regional disparities in the symbiosis of digital innovation ecosystems in the context of China is not only conducive to the coordinated development of China's regions but also to the reference and learning of other countries in the world. At present, research in this area is far from sufficient. Third, in the digital age, spatial correlation has become an important factor affecting the development of regional digital innovation ecosystems. However, there is still a lack of comprehensive investigations on how spatial correlation can promote the symbiosis of regional digital innovation ecosystems to achieve spatial convergence and what factors affect spatial convergence. On the basis of the above analysis, this paper addresses the following three key questions: (i) What is the symbiotic level of China's digital innovation ecosystem? (ii) Are there regional disparities in the symbiotic level across China's digital innovation ecosystem? (iii) What are the spatial convergence characteristics of the symbiotic level of China's digital innovation ecosystem? The answers to the abovementioned questions will have important theoretical value and practical significance for exploring countermeasures and suggestions to improve the symbiotic level of China's digital innovation ecosystem and promote high-quality economic development.

Therefore, from the perspective of symbiosis, in this paper, we take 30 provinces (cities and autonomous regions) in China from 2013 to 2022 as research samples. First, a measurement index system of the symbiotic level of regional digital innovation ecosystems is constructed from the three dimensions of digital innovation subject symbiosis, digital innovation environment symbiosis, and digital innovation interaction symbiosis, and the entropy weight TOPSIS method is applied to measure the comprehensive score of the symbiotic level of digital innovation ecosystems in various regions of China. Second, this paper applies the arithmetic mean and Dagum Gini coefficient decomposition to further analyze the overall status and regional differences in the symbiotic level of China's digital innovation ecosystem. Third, we empirically investigate the spatial convergence of the symbiotic level of China's digital innovation ecosystem by using spatial convergence analysis. Finally, this paper proposes several relevant policy implications to provide a reference for promoting overall improvements in the symbiotic level of China's digital innovation ecosystem and the coordinated development of the symbiotic level of regional digital innovation ecosystems. This study helps explore methods for measuring regional digital innovation ecosystem symbiosis, identifies the influencing factors driving regional digital innovation ecosystem symbiosis, and provides empirical evidence for the governance of regional digital innovation ecosystems in China and developing economies.

Compared with other published literature, we take the symbiosis of regional digital innovation ecosystems as the research topic and further investigate its symbiotic measurement level, regional disparities, and spatial convergence, which is pioneering and scientific. On the one hand, this paper theoretically reveals the symbiotic nature of regional digital innovation ecosystems, expands the evolution theory of digital innovation, and further improves the theoretical system of regional innovation development and "double carbon". On the other hand, from a practical point of view, this paper explores ways to improve the symbiotic level of regional digital innovation ecosystems and the diversified paths to

narrow the gap between regions, which provides useful guidance for the decision-making of relevant departments in China. The most surprising finding of this paper is that the regional economic structure and innovative application can accelerate the symbiotic spatial convergence of China's digital innovation ecosystem to a certain extent. These findings are important additions to the field globally.

The rest of this paper is organized as follows. Section 2 presents an analysis of the research status of digital innovation ecosystems. Section 3 presents the research design, including the index system construction of the symbiotic level of regional digital innovation ecosystems, methods, and data sources. Section 4 presents the empirical results and discussion of the symbiotic level of China's digital innovation ecosystem. In Section 5, the conclusions, some policy implications, and limitations are discussed.

2. Literature Review

In the global open innovation environment, promoting the development of regional innovation ecosystems is the core driving force for countries to transform old and new driving forces and achieve coordinated regional development, innovative national strategies, and high-quality economic development [28]. Moreover, the digital economy is inherently "green" in terms of technology and concept, and the information technology innovation triggered by it has accelerated the transformation and upgrading of the regional economic structure, which coincides with the requirements of the high-quality development of regional innovation ecosystems [29]. Therefore, in the process of creating new advantages in the development of the digital economy, the regional digital innovation ecosystem, which is composed of digital innovation subjects, the environment, networks, etc., can provide organizations and resource guarantees for regional digital innovation and is the key support for achieving the high-quality development of the regional economy [30,31]. In particular, with the application of symbiosis theory in the field of social sciences, promoting the symbiosis of regional digital innovation ecosystems has gradually become the only way to drive high-quality economic development with innovation [32,33].

Scholars have conducted relatively little research on the symbiotic level of digital innovation ecosystems, mainly focusing on defining the connotations, research objects, and mechanisms of digital innovation ecosystems. First, in terms of defining connotations, there is no consensus on the interpretation of the connotations of digital innovation ecosystems; rather, a trend of "let a hundred schools of thought contend" has emerged [34]. From the perspective of value cocreation, Suseno et al. (2018) defined the digital innovation ecosystem as a complex economic structure formed by individuals and organizations in the system using digital technology and participating in innovative products and services [35]. From the perspective of complex networks, Chae (2019) noted that the digital innovation ecosystem is a self-organizing system that interacts with and dynamically evolves through a variety of elements, such as organizations, individuals, tools, and environments [36]. Wang (2021) suggested that the digital innovation ecosystem is a collection of independent actors with loose coupling, collaborative interaction, and innovation and development that are under the influence of digital technology [37]. Buhe et al. (2022) suggested that the digital innovation ecosystem is based on the deep integration of digital technology and data elements, forming a multiparty interactive innovation ecosystem with characteristics such as dissipation, self-organization, and complexity [38]. Second, in terms of research objects, Li (2022) took the digital agricultural innovation ecosystem as the research object and analyzed its constituent elements from the energy group of internal architecture, the regulatory group of external architecture, and the digital agricultural innovation habitat [39]. Taking the resilience of digital innovation ecosystems as the research object, Yang et al. (2022) investigated the impact of the governance niche configuration on their

configuration [40]. Zhao et al. (2023) took the emerging industry innovation ecosystem as the research object, used the new energy vehicle industry as an example, and explored the formation mechanism of its comprehensive advantages [41]. Ge et al. (2023) took the digital economy industry innovation ecosystem as the research object and analyzed its formation process and evolution trend from four perspectives: aggregation, community, cluster, and system [42]. Finally, in terms of the mechanism, Jing et al. (2024) used the dynamic qualitative comparative analysis (QCA) method to investigate the stimulation and influence of digital innovation ecosystems on regional innovation capacity [43]. Li et al. (2023) used the fuzzy-set qualitative comparative analysis (fsQCA) method to study the configuration path driving the development of regional digital innovation ecosystems [44]. Sun et al. (2023) discussed the evolution direction of digital innovation ecosystems from the perspective of relationship interaction, knowledge ability, and behavior norms [45]. Wei et al. (2021) revealed the governance mechanism of digital innovation ecosystems from the perspective of relationship mechanisms, incentive mechanisms, and control mechanisms [46]. Taking the HUAWEI automobile as a case study, He et al. (2024) analyzed the construction mechanism and path of the core enterprise logarithm intelligent innovation ecosystem [47].

It can be seen from the literature published both at home and abroad that academia has launched a relatively rich discussion around digital innovation ecosystems, laying a foundation for the present paper. However, there are still some problems that must be further discussed. (i) The selection of research objects is a concern. Scholars usually focus on resilience, value cocreation, niche suitability, and other topics related to digital innovation ecosystems, but at the regional level, combined with symbiosis theory, research on the symbiosis of regional digital innovation ecosystems is relatively insufficient. Although some scholars have explored the symbiosis mode and evolution path of digital innovation ecosystems, few studies have measured the symbiosis level of regional digital innovation ecosystems from a quantitative perspective. Given the dual requirements of China's high-quality development and new productivity and the fact that many countries worldwide are facing the challenges of maintaining economic growth and mitigating environmental degradation [48], it is necessary to study the symbiotic level of regional digital innovation ecosystems. (ii) The perspective of regional coordinated development should be considered. Coordinating regional development has always been a major issue. Owing to the disparities in resource endowment, the economic environment, and the industrial structure among regions, the spatial–temporal evolution of the symbiotic level of regional digital innovation ecosystems is significantly different. However, few scholars have discussed regional disparities in the symbiotic level of China's digital innovation ecosystem, which is not conducive to the continuous promotion of the "Digital China" initiative. Therefore, it is necessary to further investigate regional disparities through an analysis of the overall situation of the symbiotic level of China's digital innovation ecosystem. (iii) Studying the perspective of spatial correlation is another challenge. In the past, scholars usually studied the regional digital innovation ecosystem from the perspective of classical econometrics but ignored the spatial correlation between regions, which may bias the research conclusions. Moreover, many studies have shown that China's digital innovation ecosystem objectively has certain spatial relevance. Therefore, it is necessary to further investigate the spatial convergence of the symbiotic level of China's digital innovation ecosystem and its influencing factors from the perspective of spatial correlation.

Based on the above analysis, the innovations and contributions of this paper are reflected in the following three main aspects. (i) From the perspective of symbiosis, this paper takes the symbiotic level of regional digital innovation ecosystems as the research object and constructs an indicator system for measuring the symbiotic level of regional digital

innovation ecosystems from the three dimensions of digital innovation subject symbiosis, digital innovation environment symbiosis, and digital innovation interaction symbiosis. On this basis, the entropy weight TOPSIS method with objective assignment is used to measure the comprehensive score of the symbiotic level of regional digital innovation ecosystems in China. (ii) Considering the dual requirements of overall improvement and coordinated development, this paper applies the arithmetic average and Dagum Gini coefficient decomposition to further analyze the overall status and regional disparities in the symbiotic level of China's digital innovation ecosystem to comprehensively reveal the development of the symbiotic level of China's digital innovation ecosystem. (iii) This paper empirically examines the spatial convergence characteristics of the symbiotic level of China's digital innovation ecosystem using spatial convergence analysis. Furthermore, from the perspective of economic structure and innovation application, this paper further explores the main factors affecting the spatial convergence of the symbiotic level of China's digital innovation ecosystem.

3. Research Design

3.1. Index System Construction of the Symbiotic Level of Regional Digital Innovation Ecosystems

Before the index system of the symbiotic level of regional digital innovation ecosystems is built, the first step is to clarify the meaning of regional digital innovation ecosystem. Combined with the views of previous scholars [49–52], we suggest herein that the regional digital innovation ecosystem is a complex dynamic adaptive system formed by the interaction and coevolution of digital innovation subjects and the digital innovation environment through a digital innovation interaction system. Therefore, regional digital innovation ecosystem symbiosis includes three parts: digital innovation subject symbiosis, digital innovation environment symbiosis, and digital innovation interaction symbiosis.

(1) Digital innovation subject symbiosis. From the perspective of digital innovation ecosystems, the symbiosis of subject elements is the foundation and core [53]. Previous scholars have applied diverse symbiotic populations, such as enterprises, universities, and research institutes, to represent the symbiosis of digital innovation subjects. However, they have ignored the dominance of symbiotic populations. Based on previous studies, this paper combines regional ecosystem theory to measure digital innovation subject symbiosis from the perspective of the diversity and dominance of symbiotic populations. Specifically, we select the number of industrial enterprises above the designated size, the number of colleges and universities, and the number of research institutions to evaluate the diversity of symbiotic populations. On the one hand, we select the proportion of enterprises with R&D institutions, the number of technology incubators, the number of national university science and technology parks, the average output value of high-tech industrial development zones, and the average output value of characteristic industrial bases to evaluate the dominance of symbiotic populations. Finally, eight specific indicators are used to measure digital innovation subject symbiosis.

(2) Digital innovation environment symbiosis. Owing to the self-growth and dynamic nature of digital innovation, new products and services automatically carry out iterative updates and service upgrades in combination with environmental changes. A good digital innovation environment guarantees not only the smooth progress of digital innovation activities but also the survival of symbiotic populations. In this paper we comprehensively measure digital innovation environment symbiosis in five dimensions: the economic environment, technology environment, cultural environment, opening-up environment, and financial environment. Specifically, per capita GDP and household consumption levels are selected to evaluate the economic environment;

internet broadband access ports and the trading volume of the technology market are selected to evaluate the technology environment; the population with a college degree or above and the number of books in public libraries are selected to evaluate the cultural environment; foreign technology imports and the amount of foreign investment actually utilized are selected to evaluate the opening-up environment; and the sum of the deposit and loan balances of financial institutions are selected to evaluate the financial environment. Finally, nine specific indicators are used to measure digital innovation environment symbiosis.

(3) Digital innovation interaction symbiosis. Digital innovation interaction symbiosis mainly includes a symbiotic matrix and symbiotic network, which reflect the interactions between symbiotic populations and between symbiotic populations and the symbiotic environment. The symbiotic matrix is the common resource of the symbiotic population. Owing to their different attributes and limited resources, symbiotic populations must cooperate with each other to form a symbiotic network to promote the flow of material, capital, and knowledge among innovation communities and innovation environments. Specifically, to evaluate the symbiotic matrix, total investment fixed assets, the full-time equivalent of R&D personnel, and the intramural expenditure on R&D are selected. Moreover, the proportion of government funds in enterprise R&D funds, the proportion of enterprise funds in scientific research institute funds, the proportion of government funds in the scientific and technological activity funds of scientific research institutes, the proportion of enterprise funds in university funds, the proportion of government funds in university funds, and the number of papers written by the author in cooperation with different units in the province to evaluate the symbiotic network are selected. Finally, nine specific indicators are used to measure digital innovation interaction symbiosis.

In summary, on the basis of the three subsystems of digital innovation subject symbiosis, digital innovation environment symbiosis, and digital innovation interactive symbiosis, in this paper, we construct a new measurement index system for the symbiotic level of regional digital innovation ecosystems and use the entropy weight TOPSIS method to measure the symbiotic level of China's digital innovation ecosystem. The indicator system is shown in Table 1.

Table 1. Measurement index system for the symbiotic level of regional digital innovation ecosystems.

First-Level Indicators	Second-Level Indicators	Third-Level Indicators
Digital innovation subject symbiosis	The diversity of symbiotic populations	Number of industrial enterprises above designated size
		Number of colleges and universities
		Number of research institutions
	The dominance of symbiotic populations	Proportion of enterprises with R&D institutions
		Number of technology incubators
		Number of national university science and technology parks
		Average output value of high-tech industrial development zones
		Average output value of characteristic industrial bases

Table 1. *Cont.*

First-Level Indicators	Second-Level Indicators	Third-Level Indicators
Digital innovation environment symbiosis	Economic environment	Per capita GDP
		Household consumption level
	Technology environment	Internet broadband access ports
		Trading volume of technology market
	Cultural environment	Population with college degree or above
		Number of books in public libraries
	Opening-up environment	Foreign technology imports
		Amount of foreign investment utilized
	Financial environment	Sum of deposit and loan balances of financial institutions
Digital innovation interaction symbiosis	Symbiotic matrix	Total investment fixed assets
		Full-time equivalent of R&D personnel
		Intramural expenditure on R&D
	Symbiotic network	Proportion of government funds in enterprise R&D funds
		Proportion of enterprise funds in scientific research institute funds
		Proportion of government funds in scientific and technological activity funds of scientific research institutes
		Proportion of enterprise funds in university funds
		Proportion of government funds in university funds
		Number of papers written by the author in cooperation with different units in the province

3.2. Research Methods

3.2.1. Entropy Weight TOPSIS Method

To construct a symbiosis level measurement index system for regional digital innovation ecosystems, selecting the appropriate weighting method is highly important. Compared with the weighting method of subjective judgment, the traditional entropy weight method completely depends on the data characteristics to determine the weight of each index, which can effectively avoid biased estimation caused by subjective factors. It is a more objective weighting method that has been widely used by scholars. However, the shortcomings of the traditional entropy weight method are also obvious. In other words, it considers only cross-sectional factors and ignores the consideration of time, so it is unable to compare results across different years. Therefore, we chose the entropy weight TOPSIS method to measure the symbiotic level of China's digital innovation ecosystem. The entropy weight TOPSIS method is a comprehensive evaluation method that combines the entropy weight method and the TOPSIS method. It can make full use of the information from the original data and reduce the deviation caused by subjective evaluation. Its results can objectively reflect the gap between the evaluation schemes. The steps utilized are as follows.

The first step is standardization. The evaluated object, which herein consists of 30 regions, is assumed to be $i = (1, 2, \cdots, m)$, and the measurement index is $j = (1, 2, \cdots, n)$. X_{ij} represents the data corresponding to the j index of the i region. Owing to the disparities in the magnitude and dimension of the measurement indicators, to eliminate their impact,

it is necessary to standardize X_{ij} via the range method, and the new data after processing are denoted as X'_{ij}. The formula is as follows:

$$X'_{ij} = \frac{X_{ij} - \min X_{ij}}{\max X_{ij} - \min X_{ij}} \tag{1}$$

The second step is to determine the weight value. Because the importance of each indicator is different, the entropy weight method is used to weight each indicator. The larger the weight coefficient ω_j is, the greater the importance of the measurement indicator and the greater the impact on the measurement results. The formula is as follows:

(i) Calculate the information entropy E_j of index j:

$$E_j = -\frac{1}{\ln m}\sum_{i=1}^{m}\left(\frac{X'_{ij}}{\sum_{i=1}^{m} X'_{ij}}\right)\ln\left(\frac{X'_{ij}}{\sum_{i=1}^{m} X'_{ij}}\right) \tag{2}$$

(ii) Calculate the weight value ω_j of the j index:

$$\omega_j = \frac{1 - E_j}{\sum_{j=1}^{n}(1 - E_j)} \tag{3}$$

The third step is to determine the weighted decision matrix. According to Formulas (1) and (3), the weighted decision matrix Z can be obtained as follows:

$$Z = \begin{pmatrix} z_{11} & \cdots & z_{1n} \\ \vdots & \ddots & \vdots \\ z_{m1} & \cdots & z_{mn} \end{pmatrix} \tag{4}$$

where $z_{ij} = \omega_j X'_{ij}$.

The fourth step is to calculate the comprehensive measurement of the symbiotic level. According to the weighted decision matrix, the positive ideal solution $Z_j^+ = \max(z_{1j}, z_{2j}, \cdots, z_{mj})$ and the negative ideal solution $Z_j^- = \min(z_{1j}, z_{2j}, \cdots, z_{mj})$ can be obtained. The distances D_i^+ and D_i^- between the standardized weighted decision matrix and the positive ideal solution and the negative ideal solution can be calculated. The formula is as follows:

$$D_i^+ = \sqrt{\sum_{j=1}^{n}\left(z_{ij} - Z_j^+\right)^2} \tag{5}$$

$$D_i^- = \sqrt{\sum_{j=1}^{n}\left(z_{ij} - Z_j^-\right)^2} \tag{6}$$

According to Formulas (5) and (6), the comprehensive measure of the symbiotic level in the region can be calculated. The larger the value is, the closer the measurement value of the region is to the positive ideal solution, the farther the measurement value of the region is from the negative ideal solution, and the higher the symbiotic level of its digital innovation ecosystem is. The formula is as follows:

$$C_i = \frac{D_i^-}{D_i^+ + D_i^-} \tag{7}$$

3.2.2. Dagum's Gini Coefficient Decomposition Method

To calculate the symbiotic level of digital innovation ecosystems in various regions of China using the entropy weight TOPSIS method, this paper further applies the Dagum Gini coefficient decomposition method to analyze regional disparities in the symbiotic level of China's digital innovation ecosystem. Because the Dagum Gini coefficient decomposition method compensates for the problem of overlapping data that other methods cannot examine when measuring regional disparities, it can better identify the sources of regional disparities; thus, it has become the main method for analyzing regional disparities. Moreover, this method can decompose the Gini coefficient into an intragroup coefficient G_w, an intergroup coefficient G_b, and a supervariable density coefficient G_t, which reflect the gap between the internal level of each region, the gap between the levels of each region, and the overlapping phenomenon of each region, respectively, reflecting the situation of the relative gap, and can more objectively reflect the regional differences in the symbiotic level of China's digital innovation ecosystem. First, the calculation formula for the total Gini coefficient is as follows:

$$G = \frac{\sum_{h=1}^{k}\sum_{r=1}^{k}\sum_{p=1}^{n_h}\sum_{q=1}^{n_r}\left|y_{hp} - y_{rq}\right|}{2n^2\overline{y}} \tag{8}$$

where G represents the total Gini coefficient; k represents the number of regions, k = 4; n represents the total number of provinces, $n = 30$; n_h and n_r represent the number of provinces within region h and within region r, respectively; y_{hp} represents the symbiotic level of the digital innovation ecosystem in province p within region h; y_{rq} represents the symbiotic level of the digital innovation ecosystem in province q within region r; and $\overline{y}$ represents the average symbiotic level of the digital innovation ecosystem in all provinces.

In accordance with the Dagum Gini coefficient decomposition method, the total Gini coefficient G is decomposed into the contribution rate G_w of internal disparities within each region, the contribution rate G_b of disparities between regions, and the contribution rate G_t of hyperdensity. The relationship between the four is $G = G_w + G_b + G_t$. The specific formula is as follows:

$$G_w = \sum_{h=1}^{k} G_{hh} c_h s_h \tag{9}$$

$$G_b = \sum_{h=2}^{k}\sum_{r=1}^{h-1} G_{hr}(c_h s_r + c_r s_h) D_{hr} \tag{10}$$

$$G_t = \sum_{h=2}^{k}\sum_{r=1}^{h-1} G_{hr}(c_h s_r + c_r s_h)(1 - D_{hr}) \tag{11}$$

where $c_h = \frac{n_h}{n}$ represents the ratio of the number of provinces in region h to the total number of provinces in the country; $s_h = \frac{n_h \overline{y}_h}{n\overline{y}}$, $G_{hh} = \frac{\sum_{p=1}^{n_h}\sum_{q=1}^{n_h}\left|y_{hp} - y_{hq}\right|}{2n^2\overline{y}_h}$, $G_{hr} = \frac{\sum_{p=1}^{n_h}\sum_{q=1}^{\ln}\left|y_{hp} - y_{rq}\right|}{n_h n_r(\overline{y}_h + \overline{y}_r)}$, and $h = 1, 2, \cdots k$; and D_{hr} represents the relative impact of the symbiotic level between regions. The calculation formula is as follows:

$$D_{hr} = \frac{d_{hr} - b_{hr}}{d_{hr} + b_{hr}} \tag{12}$$

where d_{hr} represents the disparity in the symbiotic levels between region h and region r, and where b_{hr} is a hypervariable first-order moment, which represents the mathematical expectation of the sum of the sample values in the regions that satisfy $y_{hp} - y_{rq} > 0$:

$$d_{hr} = \int_0^{\infty} dF_h(y) \int_0^{y} (y - x) dF_r(x) \tag{13}$$

$$b_{hr} = \int_0^{\infty} dF_r(y) \int_0^{y} (y - x) dF_h(x) \tag{14}$$

where $F_h(x)$ and $F_r(x)$ are the cumulative density distribution functions of regions h and r, respectively.

3.2.3. Spatial Convergence Analysis

(1) Spatial correlation test. The global Moran index I is used to test whether the symbiotic level of China's digital innovation ecosystem has spatial correlation characteristics. The calculation formula is as follows:

$$I = \frac{\sum_{i=1}^{m} \sum_{u=1}^{m} W_{iu}(y_i - \overline{y})(y_u - \overline{y})}{S^2 \sum_{i=1}^{m} \sum_{u=1}^{m} W_{iu}} \tag{15}$$

where $\overline{y} = \frac{1}{m} \sum_{i=1}^{m} y_i$, $S^2 = \frac{1}{m} \sum_{i=1}^{m} (y_i - \overline{y})^2$, $W_{iu} = \left\{ \begin{array}{l} \frac{1}{d_{iu}},\ i \neq u \\ 0,\ i = u \end{array} \right\}$, y_i and y_u represent the symbiotic levels of the digital innovation ecosystem in region i and region u, respectively; m represents the total number of regions; and W_{iu} represents the spatial weight matrix of geographical distance.

(2) σ convergence. The degree of dispersion is an important index for testing σ convergence. If the discrete degree of the symbiotic level of digital innovation ecosystems tends to decrease over time, this indicates that it has σ convergence. Referring to the practice of Yang et al. [54], the coefficient of variation was selected to test convergence. The calculation formula is as follows:

$$CV = S / \overline{y} \tag{16}$$

where CV represents the coefficient of variation, S represents the standard deviation of the symbiotic level of the digital innovation ecosystem, and $\overline{y}$ represents the mean value.

(3) β convergence. β convergence means that the regions with lower symbiotic levels in the initial digital innovation ecosystem have faster growth rates than the regions with higher symbiotic levels, and they finally reach a convergence state. It can also be divided into two types: absolute β convergence and conditional β convergence. Absolute β convergence means that the symbiotic level of the digital innovation ecosystem in each region has an equal change trend and eventually converges to the same level. Conditional β convergence refers to the influence of individual and structural characteristics and other factors, and the symbiotic level of the digital innovation ecosystem in each region will eventually converge to its respective stable level.

Considering that the symbiotic level of China's digital innovation ecosystem may have significant spatial correlation, this paper applies a spatial econometric model. The spatial lag model (SLM), spatial error model (SEM), and spatial Durbin model (SDM) are

commonly used spatial econometric models. Therefore, the absolute β convergence model constructed in this paper is as follows:

$$\ln\left(\frac{y_{i,t+1}}{y_{i,t}}\right) = \alpha + \beta \ln(y_{i,t}) + \rho W_{iu} \ln\left(\frac{y_{i,t+1}}{y_{i,t}}\right) + \varepsilon_{i,t} \tag{17}$$

$$\ln\left(\frac{y_{i,t+1}}{y_{i,t}}\right) = \alpha + \beta \ln(y_{i,t}) + \lambda W_{iu} \mu_{i,t} + \varepsilon_{i,t} \tag{18}$$

$$\ln\left(\frac{y_{i,t+1}}{y_{i,t}}\right) = \alpha + \beta \ln(y_{i,t}) + \rho W_{iu} \ln\left(\frac{y_{i,t+1}}{y_{i,t}}\right) + \theta W_{iu} \ln(y_{i,t}) + \varepsilon_{i,t} \tag{19}$$

where y represents the symbiotic level of the digital innovation ecosystem in each region, i represents 30 regions, t represents time, and β represents the convergence coefficient. If $\beta < 0$ and passes the significance test, then the symbiotic level of the regional digital innovation ecosystem presents a convergence trend, and vice versa. ρ represents the spatial lag coefficient, reflecting the impact of the symbiotic level growth rate of adjacent areas on the region; λ represents the spatial error coefficient, reflecting the spatial effect of the random interference term; θ represents the spatial lag coefficient of the independent variable, reflecting the influence of the symbiotic level in the adjacent area; W_{iu} represents the spatial weight matrix; and $\varepsilon_{i,t}$ and $\mu_{i,t}$ represent random disturbance terms. On the basis of the sample data, Models (17)–(19) were tested to select the most suitable model for analysis.

Conditional β convergence reflects that, owing to the different characteristics of each region, the corresponding steady level is also different, and the symbiotic level of regional digital innovation ecosystems tends to reach its own steady level over time. In fact, the initial value of the symbiotic level in various regions is only one of the factors that affects its change. On the basis of the absolute β convergence model, this paper adds some conditional variables that have an impact on the symbiotic level of the regional digital innovation ecosystem. The constructed condition β convergence model is as follows:

$$\ln\left(\frac{y_{i,t+1}}{y_{i,t}}\right) = \alpha + \beta \ln(y_{i,t}) + \rho W_{iu} \ln\left(\frac{y_{i,t+1}}{y_{i,t}}\right) + \gamma \ln H_{i,t} + \varepsilon_{i,t} \tag{20}$$

$$\ln\left(\frac{y_{i,t+1}}{y_{i,t}}\right) = \alpha + \beta \ln(y_{i,t}) + \gamma \ln H_{i,t} + \lambda W_{iu} \mu_{i,t} + \varepsilon_{i,t} \tag{21}$$

$$\ln\left(\frac{y_{i,t+1}}{y_{i,t}}\right) = \alpha + \beta \ln(y_{i,t}) + \rho W_{iu} \ln\left(\frac{y_{i,t+1}}{y_{i,t}}\right) + \theta W_{iu} \ln(y_{i,t}) + \gamma \ln H_{i,t} + \varphi W_{iu} \ln H_{i,t} + \varepsilon_{i,t} \tag{22}$$

where H represents the set of control variables. Additionally, on the basis of the sample data, the abovementioned models are tested to select the most appropriate conditional convergence model for analysis.

3.3. Data Sources and Processing

To ensure the continuity of this study, the panel data of 30 provinces in China (Xizang, Hong Kong, Macao, and Taiwan are ignored because of missing data) from 2013 to 2022 are selected as the research sample. The research data are drawn from the *China Statistical Yearbook*, *China Science and Technology Statistical Yearbook*, *China Torch Statistical Yearbook*, *China Regional Innovation Capacity Evaluation Report*, and local statistical yearbooks. The data processing of specific indicators is as follows.

First, the measurement data of most indicators are directly available from yearbooks or reports. Specifically, the number of industrial enterprises above the designated size, per

capita GDP, internet broadband access ports, the trading volume of the technology market, the population with a college degree or above, the number of books in public libraries, foreign technology imports, the amount of foreign investment utilized, total investment fixed assets, the full-time equivalent of R&D personnel, and the intramural expenditure on R&D are drawn from the *China Statistical Yearbook*. The number of colleges and universities, the number of research institutions, and the number of technology incubators are drawn from the *China Science and Technology Statistical Yearbook*. The number of national university science and technology parks is drawn from the *China Torch Statistical Yearbook*. The household consumption level and the number of papers written by the author in cooperation with different units in the province are drawn from the *China Regional Innovation Capacity Evaluation Report*. The sum of the deposit and loan balances of financial institutions is drawn from local statistical yearbooks. Second, the measurement data of some indicators need to be obtained by calculation. In the *China Science and Technology Statistical Yearbook*, the proportion of enterprises with R&D institutions is equal to the number of firms with R&D organizations divided by the number of firms above the designated size. The average output value of high-tech industrial development zones is equal to the total output value of high-tech industrial development zones divided by the number of zones. The proportion of government funds in enterprise R&D funds is equal to the government funding divided by the internal expenditure on R&D funding of enterprises above the designated size. The proportion of enterprise funds in scientific research institute funds is equal to the corporate funding divided by internal expenditures on R&D funding for research and development organizations. The proportion of government funds in scientific and technological activity funds of scientific research institutes is equal to the government funding divided by internal expenditure on R&D funding for research and development organizations. The proportion of enterprise funds in university funds is equal to corporate funding divided by internal expenditure on R&D funding for higher education institutions. The proportion of government funds in university funds is equal to the government funding divided by internal expenditure on R&D funding for higher education institutions. In the *China Torch Statistical Yearbook*, the average output value of characteristic industrial bases is equal to the total output value of specialty industrial bases divided by the number of bases. Third, in the process of data collection, individual data are still missing. To ensure the consistency of the statistical caliber, this paper adopts linear fitting and exponential smoothing to supplement them.

4. Empirical Analysis

4.1. Analysis of the Results of the Symbiosis of Digital Innovation Ecosystems

The entropy weight TOPSIS method is used to calculate the symbiotic level of the digital innovation ecosystems in 30 provinces of China from 2013 to 2022. Moreover, the 30 provinces are further divided into eastern, middle, western, and northeast regions to more intuitively present the regional disparities and spatial distributions in the symbiotic level of China's digital innovation ecosystem. The measurement results are shown in Tables 2 and 3.

According to the measurement results in Tables 2 and 3, the following findings are obtained.

First, at the national level, on the one hand, the average annual value of the symbiotic level of China's digital innovation ecosystem is 0.187, which is lower than the average symbiotic level of the digital innovation ecosystem in the eastern region and higher than the levels of the other three regions. There are 12 provinces whose average is higher than the national average and 18 provinces whose average is lower than the national average. This suggests that the symbiotic level of the digital innovation ecosystem in 60% of the provinces is still weak. On the other hand, the average symbiotic level of China's digital innovation ecosystem increased from 0.167 in 2013 to 0.223 in 2022, with an average annual growth rate

of 3.73%. This result shows that the symbiotic level of China's digital innovation ecosystem is on the rise and that its future development looks promising.

Table 2. Measurement results of the symbiotic level of the digital innovation ecosystem in various provinces of China.

Provinces	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	Annual Average	Ranking
Beijing	0.329	0.327	0.344	0.352	0.393	0.401	0.493	0.462	0.488	0.507	0.410	3
Tianjin	0.192	0.196	0.197	0.183	0.205	0.136	0.149	0.142	0.158	0.165	0.172	14
Hebei	0.109	0.120	0.127	0.135	0.144	0.156	0.172	0.182	0.196	0.208	0.155	16
Shanxi	0.084	0.083	0.086	0.091	0.106	0.117	0.090	0.118	0.116	0.122	0.101	20
Inner Mongolia	0.099	0.086	0.080	0.084	0.078	0.079	0.083	0.077	0.085	0.092	0.084	27
Liaoning	0.225	0.223	0.151	0.143	0.143	0.145	0.154	0.154	0.162	0.170	0.167	15
Jilin	0.095	0.103	0.098	0.110	0.087	0.090	0.172	0.087	0.086	0.087	0.101	21
Heilongjiang	0.136	0.143	0.153	0.145	0.147	0.134	0.121	0.112	0.119	0.121	0.133	19
Shanghai	0.328	0.307	0.283	0.303	0.326	0.321	0.388	0.352	0.427	0.436	0.347	4
Jiangsu	0.407	0.388	0.424	0.403	0.426	0.441	0.486	0.483	0.536	0.553	0.455	2
Zhejiang	0.276	0.233	0.249	0.273	0.299	0.314	0.346	0.361	0.413	0.438	0.320	5
Anhui	0.133	0.147	0.160	0.167	0.180	0.193	0.206	0.222	0.261	0.267	0.193	11
Fujian	0.124	0.144	0.128	0.148	0.148	0.148	0.156	0.163	0.192	0.188	0.154	17
Jiangxi	0.090	0.099	0.110	0.121	0.132	0.145	0.163	0.175	0.192	0.180	0.141	18
Shandong	0.225	0.239	0.252	0.263	0.280	0.296	0.295	0.302	0.351	0.381	0.288	6
Henan	0.147	0.160	0.173	0.187	0.192	0.201	0.215	0.223	0.242	0.232	0.197	10
Hubei	0.138	0.151	0.166	0.181	0.188	0.199	0.220	0.215	0.256	0.278	0.199	8
Hunan	0.174	0.156	0.189	0.177	0.196	0.200	0.208	0.221	0.212	0.247	0.198	9
Guangdong	0.392	0.381	0.435	0.480	0.475	0.466	0.528	0.580	0.586	0.640	0.496	1
Guangxi	0.074	0.077	0.080	0.083	0.089	0.099	0.109	0.107	0.123	0.118	0.096	24
Hainan	0.101	0.067	0.066	0.070	0.066	0.055	0.052	0.064	0.070	0.069	0.068	30
Chongqing	0.237	0.242	0.196	0.233	0.286	0.274	0.186	0.189	0.206	0.205	0.226	7
Sichuan	0.149	0.148	0.161	0.168	0.175	0.193	0.214	0.217	0.235	0.239	0.190	12
Guizhou	0.113	0.108	0.088	0.085	0.090	0.087	0.095	0.105	0.099	0.097	0.097	23
Yunnan	0.097	0.090	0.105	0.088	0.095	0.092	0.108	0.102	0.102	0.114	0.099	22
Shaanxi	0.150	0.155	0.165	0.163	0.167	0.166	0.182	0.178	0.202	0.220	0.175	13
Gansu	0.086	0.081	0.079	0.083	0.082	0.090	0.126	0.087	0.090	0.091	0.089	26
Qinghai	0.095	0.083	0.087	0.091	0.085	0.076	0.066	0.068	0.062	0.067	0.078	28
Ningxia	0.077	0.072	0.067	0.069	0.083	0.072	0.086	0.077	0.078	0.089	0.077	29
Xinjiang	0.115	0.123	0.125	0.122	0.076	0.079	0.076	0.072	0.079	0.080	0.095	25

Table 3. Measurement results of the symbiotic level of the digital innovation ecosystem in the four regions and the entire country.

Regions	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	Annual Average	Ranking
Eastern region	0.248	0.240	0.250	0.261	0.276	0.273	0.306	0.309	0.342	0.359	0.287	1
Middle region	0.128	0.133	0.147	0.154	0.166	0.176	0.184	0.196	0.213	0.221	0.172	2
Western region	0.117	0.115	0.112	0.115	0.119	0.119	0.121	0.116	0.124	0.128	0.119	4
Northeast region	0.152	0.156	0.134	0.133	0.126	0.123	0.149	0.118	0.123	0.126	0.134	3
National region	0.167	0.164	0.167	0.173	0.181	0.182	0.198	0.197	0.214	0.223	0.187	-

Note: The eastern region includes Beijing, Tianjin, Hebei, Shanghai, Jiangsu, Zhejiang, Fujian, Shandong, Guangdong, and Hainan; the middle region includes Shanxi, Anhui, Jiangxi, Henan, Hubei, and Hunan; the western region includes Inner Mongolia, Guangxi, Chongqing, Sichuan, Guizhou, Yunnan, Shaanxi, Gansu, Qinghai, Ningxia, and Xinjiang; and the northeast region includes Liaoning, Jilin, and Heilongjiang.

Second, from the perspective of the four regions, on the one hand, the average level of digital innovation ecosystem symbiosis in the eastern region is 0.287, which is high, followed by that of the middle region and that of the western region. Moreover, the average disparity between the digital innovation ecosystem symbiotic levels of the west and northeast regions is the smallest, whereas that between other regions is relatively large; this is especially true in the eastern region, where the disparity is more than two times greater than that in the western region. On the other hand, the average level of digital innovation ecosystem symbiosis in the eastern region increased from 0.248 to 0.359, with an average annual growth rate of 4.97%. The average level of symbiosis in the middle region increased from 0.128 to 0.221, with an average annual growth rate of 8.07%. The

average level of symbiosis in the western region increased from 0.117 to 0.128, with an average annual growth rate of 1.04%. The average level of symbiosis in the northeast region decreased from 0.152 to 0.126, and the average annual growth rate was negative. These data show that the symbiotic level of the digital innovation ecosystems in China's four regions is not balanced and that there are large disparities between the regions.

Third, at the provincial level, there are great disparities in the symbiotic level of the digital innovation ecosystem among provinces in China. The top five provinces in terms of the average level of digital innovation ecosystem symbiosis are Guangdong, Jiangsu, Beijing, Shanghai, and Zhejiang, all of which are in the eastern region. The nine provinces with the lowest level of symbiosis are Yunnan, Guizhou, Guangxi, Xinjiang, Gansu, Inner Mongolia, Qinghai, Ningxia, and Hainan. Except for Hainan, all of these provinces are in the western region. Notably, the average symbiotic level of the Guangdong digital innovation ecosystem, which is ranked first, is 7.29 times that of Hainan, which is ranked thirtieth. On the one hand, the overall symbiotic level of China's digital innovation ecosystem presents a declining trend in the eastern, middle, northeast, and western regions. On the other hand, the growth rate of the symbiotic level in the middle region is the fastest, exceeding that in the eastern region and thereby forming a "catch-up effect" in that region. However, the growth rates of the western and northeast regions are low, and the gap between the symbiotic levels of the eastern and middle regions is expanding.

To further reveal the regional disparities in the symbiotic level of the digital innovation ecosystem in each province of the four regions, we use the K-means method to cluster the symbiotic level of the digital innovation ecosystem from 2013 to 2022. The results are shown in Table 4. These results suggest that the five provinces of Guangdong, Jiangsu, Beijing, Shanghai, and Zhejiang belong to the leading type of digital innovation ecosystem symbiosis area. The 12 provinces of Shandong, Hebei, Fujian, Tianjin, Hubei, Anhui, Henan, Hunan, Sichuan, Chongqing, Shaanxi, and Liaoning are next, belonging to the ordinary type of digital innovation ecosystem symbiosis area. The remaining 13 provinces belong to the lagging type of digital innovation ecosystem symbiosis area. The leading types of areas are located in the eastern region. Among the lagging areas, eight provinces are located in the western region; that is, 72.73% of the provinces in the western region are of the lagging type. This shows that the level of innovation ecosystem symbiosis in the western region is relatively lagging overall.

Table 4. Cluster distribution of the symbiotic level of the digital innovation ecosystem.

Type	Eastern Region	Middle Region	Western Region	Northeast Region
Leading type	Guangdong, Jiangsu, Beijing, Shanghai, Zhejiang			
Ordinary type	Shandong, Hebei, Fujian, Tianjin	Hubei, Anhui, Henan, Hunan	Sichuan, Chongqing, Shaanxi	Liaoning
Lagging type	Hainan	Jiangxi, Shanxi	Guangxi, Yunnan, Guizhou, Gansu, Inner Mongolia, Xinjiang, Ningxia, Qinghai	Heilongjiang, Jilin

In general, the symbiotic level of China's digital innovation ecosystem from 2013 to 2022 presents obvious characteristics of space–time differentiation, forming a spatial distribution pattern that is "high in the east, flat in the middle, and low in the west".

From the perspective of time, the symbiotic level of digital innovation ecosystems in most regions of China is increasing. Spatially, the provinces located in the leading-type

areas are in the eastern region, the provinces located in the ordinary-type areas are mainly in the middle region, and the provinces located in the lagging-type areas are mainly in the western and northeast regions. The symbiotic level of the digital innovation ecosystem in most regions of China still belongs to either the ordinary type or the lagging type. Notably, although the symbiotic level of China's digital innovation ecosystem has shown a positive development trend as a whole, the disparities between regions are further expanding over time, which is not conducive to coordinated regional development.

4.2. Regional Disparity Analysis of the Symbiotic Levels of Digital Innovation Ecosystems

To conduct a deep analysis of the overall disparities and their sources in the symbiotic level of the digital innovation ecosystem, we use the Dagum Gini coefficient decomposition method to decompose the overall disparities into disparities within a region, disparities between regions, and hypervariable density. The results are shown in Table 5.

Table 5. Dagum Gini coefficient and its decomposition results.

Year	Overall	Disparities Within the Regions				Disparities Between Regions						Contribution Rate (%)		
		East	Middle	West	Northeast	East–Middle	East–West	East–Northeast	Middle–West	Middle–Northeast	West–Northeast	Disparities Within the Regions	Disparities Between Regions	Hyper Variable Density
2013	0.290	0.251	0.135	0.191	0.191	0.360	0.393	0.316	0.182	0.191	0.227	22.580	61.691	15.729
2014	0.288	0.249	0.117	0.211	0.171	0.339	0.398	0.296	0.196	0.175	0.241	22.772	60.247	16.981
2015	0.297	0.269	0.134	0.198	0.091	0.330	0.421	0.355	0.210	0.137	0.186	22.867	62.969	14.164
2016	0.301	0.267	0.120	0.212	0.060	0.322	0.427	0.361	0.227	0.140	0.195	22.698	63.378	13.923
2017	0.318	0.263	0.106	0.240	0.106	0.321	0.447	0.404	0.264	0.177	0.218	22.073	62.659	15.269
2018	0.324	0.278	0.093	0.249	0.100	0.318	0.454	0.415	0.274	0.193	0.219	22.395	60.735	16.870
2019	0.332	0.295	0.122	0.216	0.075	0.349	0.480	0.395	0.256	0.166	0.196	22.126	64.519	13.355
2020	0.344	0.295	0.095	0.227	0.127	0.326	0.491	0.475	0.283	0.261	0.200	21.535	66.502	11.963
2021	0.353	0.279	0.123	0.245	0.137	0.329	0.503	0.495	0.301	0.285	0.216	20.995	67.322	11.683
2022	0.358	0.284	0.133	0.237	0.146	0.338	0.509	0.504	0.302	0.289	0.214	20.970	67.272	11.758
mean	0.321	0.273	0.118	0.223	0.120	0.333	0.452	0.402	0.250	0.201	0.211	22.101	63.729	14.170

Table 5 shows the following findings.

First, in terms of overall disparities, the overall Gini coefficient has steadily increased from 0.290 in 2013 to 0.358 in 2022, with an average of 0.321. This indicates that there is a significant imbalance in the symbiotic level of China's digital innovation ecosystem.

Second, at the level of disparities within the regions, the Gini coefficients of the four major regions range from 0.060 to 0.295, all of which are lower than the overall national level. Among them, the largest disparity within the region is found in the eastern region, as shown in Table 4. The provinces in the eastern region are distributed among the leading, ordinary, and lagging types, and the smallest disparity within the region is found in the northeast region. Moreover, the evolutionary trends of disparities within a region differ. Although the Gini coefficient in the eastern and western regions has slightly decreased in some years, it has shown a slow upward trend overall. The trend of change in the middle region is opposite to that found in the eastern and western regions, showing a slow downward trend. The Gini coefficient within the northeast region fluctuates greatly and shows an overall downward trend.

Third, in terms of disparities between regions, the disparities between the eastern and western regions of China are the largest, followed by those between the eastern and northeastern regions; the disparities between the middle and northeastern regions are the smallest. These results are reasonable. The eastern region of China has obvious regional advantages, economic foundation advantages, and policy advantages. It not only has more digital infrastructure and sufficient R&D investment but can also attract significant digital talent and accelerate the symbiosis of digital innovation ecosystems. The economic development of the western region has made great progress since the implementation of the western development strategy. However, owing to its weak economic foundation, it is unable to provide sufficient funds and talent support for digital innovation activities; coupled with the lack of digestion and absorption capacity, it cannot effectively use the

external knowledge brought by knowledge flow, which is not conducive to the symbiosis of the digital innovation ecosystem [55]. The middle region has better location advantages than the western region. It can make full use of the superior geographical conditions close to the developed eastern region, imitate and learn from its advanced experience, and steadily improve the symbiotic level of its digital innovation ecosystem. Although the development of the digital innovation ecosystem in the northeast region has been slow, relying on the old industrial base in Northeast China and policy preferences, it is continuously improving its symbiotic level.

Fourth, in terms of the contribution rate, the mean contribution rates of disparities within the regions, disparities between regions, and hypervariable density within the sample period are 22.101%, 63.729%, and 14.170%, respectively. The contribution rate of disparities between regions is as high as 63.729%, indicating that disparities between regions are the main reason for disparities in the symbiotic level of China's digital innovation ecosystem.

4.3. Spatial Convergence Analysis of the Symbiotic Levels of Digital Innovation Ecosystems

4.3.1. σ Convergence Analysis

In accordance with the abovementioned research methods, we use the coefficient of variation of the symbiotic level of digital innovation ecosystems to measure convergence. The coefficients of variation of the symbiotic level across the whole country and the four regions are shown in Figure 1.

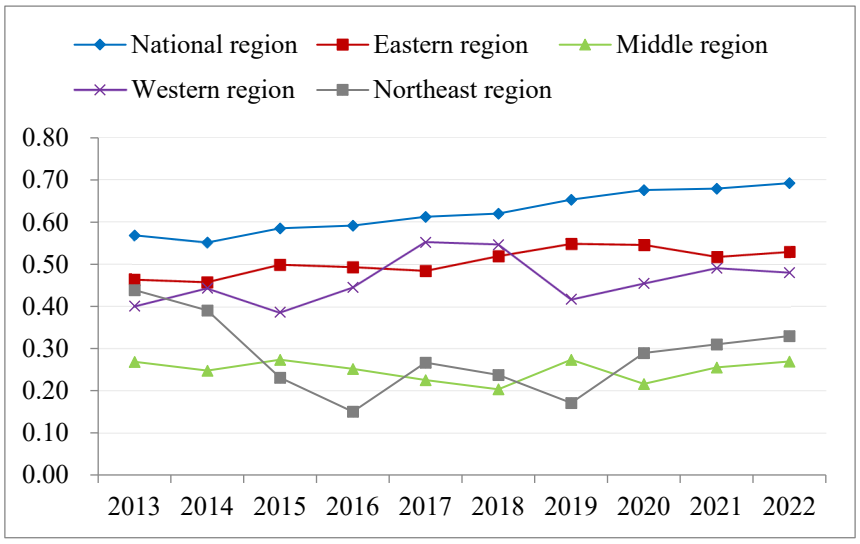


Figure 1. The coefficient of variation of the symbiotic level of digital innovation ecosystems in China and four major regions.

Figure 1 shows that at the national level, the coefficient of variation of the symbiotic level of digital innovation ecosystems increased from 0.569 in 2013 to 0.693 in 2022, which shows an overall upward trend; this indicates that regional disparities continue to expand. At the regional level, the development trend of the coefficient of variation of the symbiotic level of the innovation ecosystems in the four regions is different. Specifically, although the coefficient of variation of the eastern region fluctuates slightly, it shows a slow upward trend overall. The coefficient of variation of the western region shows an increasing trend. Although the variation coefficient of the middle region has increased and decreased, the overall situation is relatively stable. The variation coefficient of the northeast region has fluctuated greatly, showing a downward trend in the early stage and an upward trend in recent years. In general, the coefficients of variation of the symbiotic level of the innovation

ecosystems in the nation and the four regions do not show evidence of σ convergence; otherwise, such convergence is not obvious. Disparities between regions are expanding, and the regional imbalance is aggravated.

4.3.2. β Convergence Analysis

(1) Spatial correlation test. We use the global Moran index method to test the spatial correlation of the symbiotic level of China's digital innovation ecosystem. The results are shown in Table 6. The results suggest that the global Moran index of the symbiotic level of China's digital innovation ecosystem from 2013 to 2022 was significantly positive. This shows that the symbiotic level of China's digital innovation ecosystem has a significant positive spatial correlation; that is, it has the characteristics of spatial agglomeration.

Table 6. Global Moran index of the symbiotic level of the digital innovation ecosystem.

Year	Global Moran Index	Z Value	*p* Value
2013	0.009 *	1.336	0.091
2014	0.014 *	1.476	0.070
2015	0.009 *	1.364	0.086
2016	0.008 *	1.326	0.092
2017	0.017 *	1.558	0.060
2018	0.017 *	1.569	0.058
2019	0.017 *	1.588	0.056
2020	0.025 **	1.844	0.033
2021	0.047 ***	2.496	0.006
2022	0.041 **	2.297	0.011

Note: ***, **, and * represent significance at the 1%, 5%, and 10% levels, respectively.

(2) Analysis of absolute β convergence. In the spatial econometric model, the spatial correlation is reflected mainly in the lag term and the error term of the dependent variable. There are two basic spatial econometric models: the spatial lag model (SAR) and the spatial error model (SEM). The SAR model is mainly used to study the influence of the behavior of adjacent space units on other units in the whole system. The SEM reflects the relationships between spatial units through the error term. With in-depth studies, the spatial Durbin model (SDM) has been widely used by scholars. Compared with the SAR and SEMs, the SDM introduces the spatial lag term of explanatory variables and explained variables, which can address the problem of variable omission and has better stability and reliability. The commonly used spatial econometric models can be divided into three main types: SAR, SEM, and SDM. Moreover, the economic implications of the three spatial econometric models differ.

To ensure the accuracy and scientific validity of the research results, this paper applies the Hausman test, LM test, and robust LM test to screen Models (17)–(19) to select the optimal spatial model to analyze the absolute β convergence of the symbiotic level of the digital innovation ecosystems in the nation and the four examined regions. The results are shown in Table 7.

Table 7 shows the following findings.

First, the absolute convergence coefficients β of the national, eastern, western, and northeast regions are significantly negative at the 1% level. This shows that the symbiotic level of the digital innovation ecosystems in the national, eastern, western, and northeast regions presents an absolute convergence β trend. The regions with low symbiotic levels in digital innovation ecosystems have a "catch-up effect" on the regions with high symbiotic levels. In other words, in the sample period, regions with a low degree of digital innovation

ecosystem symbiosis have faster growth rates at the symbiotic level than regions with a high degree of symbiosis; however, the absolute disparity in the degree of digital innovation ecosystem symbiosis among regions has not decreased.

Table 7. Test results of absolute β convergence in China and the four examined regions.

Models	National Region	Eastern Region		Middle Region	Western Region	Northeast Region
	SDM (FE)	SAR (FE)	SEM (FE)	OLS (RE)	OLS (FE)	OLS (FE)
β	−0.395 *** (0.077)	−0.212 *** (0.060)	−0.319 *** (0.065)	−0.125 ** (0.061)	−0.352 *** (0.081)	−0.845 *** (0.210)
ρ	0.301 *** (0.110)	0.421 *** (0.125)				
λ			0.548 *** (0.101)			
θ	0.426 *** (0.100)					
R^2	0.233	0.027	0.088	0.082	0.179	0.453
Hausman test	58.480 ***	16.244 ***	29.304 ***	3.525	16.435 ***	7.716 ***
LM spatial lag	3.303 *	4.502 **	4.502 **	1.215	0.010	0.011
Robust LM spatial lag	6.329 **	0.565	0.565	0.004	0.388	2.558
LM spatial error	3.033 *	4.337 **	4.337 **	1.213	0.008	0.124
Robust LM spatial error	6.059 **	0.400	0.400	0.002	0.386	2.671
Number of samples	270	90		54	99	27

Notes: ***, **, and * represent significance at the 1%, 5%, and 10% levels, respectively; the numbers in brackets are standard errors.

Second, the spatial effects of the nation and the four regions are heterogeneous. (i) The absolute convergence model of the symbiotic level of the digital innovation ecosystem in China is the spatial dominance model. This shows that the symbiotic level of digital innovation ecosystems among regions in China has obvious spatial agglomeration effects and correlation effects. Spatial interaction will occur in adjacent or similar regions. The sharing of digital technology, the flow of economic factors, and the dissemination of cultural awareness are conducive to optimizing the rational allocation of digital resources and innovation factors, realizing knowledge sharing, generating economies of scale, and promoting the spatial agglomeration of digital innovation activities. (ii) In the eastern region, the SAR model and SEM, which have spatial lag effects and spatial error effects, can be used. This shows that the symbiotic level of the digital innovation ecosystem in the eastern region will not only be affected by the spatial spillover effect in other regions but also promote digital innovation in other regions. On the one hand, owing to its geographical advantages and strong economic strength, the eastern region can provide a good development platform for digital innovation activities, which is conducive to the introduction of advanced technology and talent. Moreover, this region has a strong ability to undertake technology transfer, which can effectively digest and absorb advanced science and technology to transform research results into market products and further promote the development of the digital economy. On the other hand, with the effective implementation of national policies, the economic level of other regions, which can effectively use superior digital innovation resources in the eastern region, has further improved. Improvements in the digestion and absorption capacity of external resources in other regions have also promoted the symbiosis level of digital innovation ecosystems to a certain extent. (iii) The central, western, and northeast regions have not passed the LM test, and there is no spatial effect. Owing to the limited geographical location and low economic level in the middle, western, and northeast regions, the ability to use digital innovation resources is weak, and

the digital innovation environment in the surrounding regions is moderate, resulting in the spatial spillover effect of these regions not being obvious [56].

(3) Analysis of conditional β convergence. To further analyze the convergence β of the symbiotic level of China's digital innovation ecosystem, we include two control variables, namely, economic structure (ES) and innovation application (IA), in the convergence model. We use the proportion of the added value of the third industry to the local GDP to represent the economic structure and use the proportion of the new product development expenditure of enterprises above the designated size to new product sales revenue to represent innovative application. The results are shown in Table 8.

Table 8. Convergence test results of conditional β in China and four major regions.

Models	National Region		Eastern Region		Middle Region	Western Region	Northeast Region
	SAR (FE)	SEM (FE)	SAR (FE)	SEM (FE)	OLS (FE)	OLS (FE)	OLS (FE)
β	−0.381 ***	−0.401 ***	−0.286 ***	−0.282 ***	−0.403 ***	−0.408 ***	−0.931 ***
	(0.072)	(0.044)	(0.072)	(0.079)	(0.107)	(0.081)	(0.182)
lnES	0.266 ***	0.185 **	0.358	0.336	0.493 ***	0.217 *	−0.216
	(0.074)	(0.107)	(0.234)	(0.292)	(0.162)	(0.110)	(0.243)
lnIA	0.092 ***	0.083 ***	0.114 ***	0.105 **	−0.024	0.072 **	0.326 **
	(0.022)	(0.024)	(0.043)	(0.047)	(0.067)	(0.031)	(0.146)
ρ	0.234 **		0.323 ***				
	(0.109)		(0.072)				
λ		0.408 ***		0.315 ***			
		(0.154)		(0.086)			
R^2	0.233	0.230	0.286	0.264	0.240	0.242	0.580
Hausman test	106.458 ***	111.422 ***	38.179 ***	34.226 ***	13.017 ***	22.495 ***	9.224 **
LM spatial lag	3.455 *	3.455 *	4.646 **	4.646 **	1.235	0.041	0.063
Robust LM spatial lag	2.901 *	2.901 *	0.000	0.000	0.988 *	1.311	2.345
LM spatial error	3.306 *	3.306 *	4.744 **	4.744 **	1.037	0.070	0.073
Robust LM spatial error	2.752	2.752	0.098	0.098	0.790	1.340	2.355
Number of samples	270		90		54	99	27

Notes: ***, **, and * represent significance at the 1%, 5%, and 10% levels, respectively; the numbers in brackets are standard errors.

Table 8 reveals the following findings.

First, the conditional convergence coefficient β for the nation and the four major regions is significantly negative. This finding indicates that the symbiotic level of the digital innovation ecosystems in the nation and the four major regions shows a conditional β convergence trend; that is, the symbiotic level of the digital innovation ecosystems in these regions converges toward their respective steady-state levels.

Second, for the national region, the regression coefficients for economic structure and innovative application are significantly positive, indicating that economic structure and innovative application can accelerate the symbiotic convergence of digital innovation ecosystems to a certain extent. On the one hand, the digital economy has improved the proportion of the tertiary industry and optimized the regional economic structure by promoting the development of digital industrialization and the digital transformation and upgrading of traditional industries. A better economic structure means the widespread allocation and layout of innovation resources, which can continuously promote the gradual transformation of China's innovation ecosystem to a digital innovation ecosystem. This has also accelerated collaborative innovation R&D and specialized division of labor between different regions, thus promoting the spatial convergence of the symbiotic level of China's digital innovation ecosystem. On the other hand, the digital economy has removed policy

barriers and protectionism, which can improve innovation exchange between regions and reduce the cost of digital innovation applications. The flow of digital talent and innovation capital across regions can promote the spatial convergence of the symbiotic level of digital innovation ecosystems in surrounding areas.

Third, disparities exist in the spatial effects across the nation and the four major regions. Among them, the convergence model for the symbiotic level of the digital innovation ecosystem in the national and eastern regions can be either the spatial lag model or spatial error model, indicating that the symbiotic level of the digital innovation ecosystems in the national and eastern regions has significant spatial correlation and that improvements in the symbiotic level in surrounding areas have a significant promoting effect on local symbiosis. In addition, the middle, western, and northeastern regions did not pass the LM test, which means that there was no spatial effect. The possible reason for this result is that, compared with the eastern region, the middle, western, and northeastern regions are relatively weak in terms of economic foundation, innovation transformation, and application, which is not conducive to the coordinated and sustainable development of China's digital innovation ecosystem.

5. Concluding Remarks

5.1. Conclusion and Implications

This paper estimates and analyzes the symbiotic level, regional disparities, and spatial convergence of digital innovation ecosystems in 30 provinces of China from 2013 to 2022. The main conclusions are as follows.

(1) At the national level, the symbiotic level of China's digital innovation ecosystem is generally on the rise, and its future development looks promising. Moreover, the symbiotic level of China's digital innovation ecosystem presents obvious space–time differentiation characteristics, forming a spatial distribution pattern that is "high in the east, flat in the middle, and low in the west". The policy implications are as follows. The Chinese government should seize this development opportunity to further improve the symbiosis level of digital innovation ecosystems by improving the collaborative symbiosis development mechanism of digital innovation subjects. First, the Chinese government should continue to promote innovation-driven development strategies. By giving full play to the leading role of large enterprises, colleges and universities, scientific research institutions, and other digital innovation entities, the Chinese government can guide more small and medium-sized enterprises, social capital, and other entities to cooperate in digital innovation and jointly improve the symbiosis level of the digital innovation ecosystem. Second, the Chinese government should effectively expand the cooperation width and breadth of digital innovation. The digital innovation ecosystem in each region should aim to strengthen the symbiosis of elements within the region. The government should establish a full element symbiosis center that includes digital innovation subjects, as well as digital innovation subjects and the digital innovation environment, to promote the deep integration of "government, industry, academia, research, and application", enhance the level of element symbiosis, and release the driving force of high-quality innovation in the region empowered by elements. Third, each regional digital innovation ecosystem should attach importance to factor endowment, uphold the innovation concept of open communication, and explore its own path to improve the symbiotic level of the digital innovation ecosystem according to local conditions. Through an innovation technology alliance, the government can build a cooperation platform for the flow of elements between various regions, establish a high-end systematic symbiotic network of digital innovation, promote the effective spatial allocation of digital inno-

vation resources, and enhance the ability of elements to enable the development of digital innovation.

(2) From a regional perspective, excessive disparities between regions are the primary factors contributing to the overall difference in the symbiotic level of China's digital innovation ecosystem. The policy implications are as follows. The Chinese government should build a regional hierarchical development system to realize the symbiotic and coordinated development of the digital innovation ecosystem among regions. First, the symbiotic level of the digital innovation ecosystem in the eastern region is always higher than that in the other three regions and shows a significant growth trend. Therefore, the eastern region can increase scientific research investment, enhance the wide application of digital technologies such as big data, blockchain, and artificial intelligence in multiple scenarios, constantly innovate the business model of digital technology, accelerate the transformation of the quality and efficiency of the digital economy, and continuously improve the symbiosis level of its own digital innovation ecosystem. Second, for the middle, western, and northeast regions, on the one hand, the government can eliminate the development barriers of digital innovation resource allocation by means of a rational allocation of digital innovation elements, improvements in the resource management system and mechanism, and the creation of a science and technology market system. On the other hand, the government can build a regional digital innovation development system with development dislocation and policy mutual assistance and provide research funding and digital talent support to improve the symbiotic level of the digital innovation ecosystem.

(3) From the perspective of σ convergence, regional disparities at the symbiotic level of digital innovation ecosystems are expanding, and uneven regional development is intensifying. The policy implications are as follows. The Chinese government should narrow the gap in the symbiotic level of digital innovation ecosystems between regions and deepen the integrated development of digital innovation among regions. First, the government can build a collaborative regional digital innovation strategic layout and explore the relative balance and dynamic synergy of digital innovation development among regions. On the one hand, the government should strengthen overall planning and layout at the national level and scientifically and reasonably allocate innovative resources on the basis of the advantages and industrial development needs of each region. On the other hand, by focusing on benchmark demonstrations such as the construction of international science and technology innovation centers in Beijing, Shanghai, and the Guangdong–Hong Kong–Macao Greater Bay Area, the government should fully leverage the spillover effects of regional digital innovation cooperation networks, encourage the acceleration of the digital innovation layout in areas with network edge nodes, and form a new pattern that promotes the accelerated development of the eastern region, the rise of the middle region, the development of the western region, and the overall revitalization of the northeast. Second, the government can increase investment in digital technology, encourage regions to set up special funds for digital economic development, and reward digital innovation entities that have achieved certain results at different levels. Moreover, the government can reduce taxes and fees for digital innovation subjects who carry out digital technology innovation activities in terms of fiscal and tax policies and provide government regulation to promote the symbiosis of digital innovation ecosystems.

(4) From the perspective of absolute β convergence, regions with low symbiotic levels within digital innovation ecosystems have faster growth rates at the symbiotic level than regions with high symbiotic levels, and there is a certain spatial spillover effect. The policy implications are as follows. The Chinese government should focus on

optimizing the efficiency of the market-oriented allocation of factors, build an ecological development mechanism for marketization and the coexistence of competition and cooperation, and further promote the flow of digital symbiotic factors to less developed regions. First, relying on the national "East Counts, West Counts" project, the government can actively guide internet enterprises in the eastern region to set up new data centers in the middle, western, and northeastern regions, represented by cloud computing and big data, to give full play to the driving effect of the digital space of complementary advantages and synergistic linkages. Second, the middle region should make full use of its advantages in connecting the eastern and western regions. It should not only encourage the basic digital industries in the central region to move to the western region but also encourage the relocation of high-end digital industries from the eastern region to further promote the upgrading of the regional digital industry and achieve the symbiotic evolution of the regional digital innovation ecosystem. Third, for the western and northeastern regions, while striving to improve their own level of economic development, they should actively expand the application scenarios of digital technology, promote digital industrialization and industrial digitization around local natural resource advantages and special industries, enhance the symbiosis level of digital innovation ecosystems, and narrow the gap with the eastern and middle regions.

(5) From the perspective of conditional β convergence, the economic structure and innovative application can accelerate the symbiotic spatial convergence of China's digital innovation ecosystem to a certain extent. The policy implications are as follows. The Chinese government should optimize the economic structure and enhance our ability to transform and apply digital innovation. Each region should attach importance to the driving role of the tertiary industry in the development of digital innovation, increase the proportion of the tertiary industry in the gross domestic product, further optimize the economic structure, and promote the symbiotic development of all factors in the regional digital innovation ecosystem. Moreover, all regions should promote the transformation and implementation of digital innovation achievements, highlight and strengthen the role and cooperation of various digital innovation subjects in practical applications, shorten the gap between regions, accelerate the convergence of China's digital innovation ecosystem symbiosis, and, ultimately, achieve the high-quality development of China's digital innovation ecosystem.

5.2. Theoretical Contribution

(1) Currently, the literature on digital innovation ecosystems focuses mainly on resilience, value cocreation, ecological niche suitability, and other topics. Few studies have explored the digital innovation ecosystem at the regional management level in combination with symbiosis theory, and few studies have used multidimensional comprehensive indicators to analyze the symbiosis of digital innovation ecosystems. In this paper, the concept and connotations of regional digital innovation ecosystems are first clarified through the use of symbiosis theory. Then, a measurement index system of the symbiotic level of regional digital innovation ecosystems is constructed from the perspective of digital innovation subject symbiosis, digital innovation environment symbiosis, and digital innovation interaction symbiosis, and a comprehensive evaluation of the overall effect of the regional digital innovation ecosystem is conducted. This study further improves the theoretical system of digital innovation ecosystems.

(2) Coordinated regional development has always been a major issue. Previous studies on China's digital innovation ecosystems have focused mostly on the endogenous operation mechanism and evolution law. Meanwhile, research on digital innovation ecosystems in

other countries is based mainly on the industrial perspective. Research on disparities in the symbiotic level of digital innovation ecosystems in different regions is lacking. This paper focuses on the regional disparities in the symbiotic level of digital innovation ecosystems in China, which not only theoretically broadens the research perspective of digital innovation ecosystems but also provides practical guidance on how China and other countries can gain regional competitive advantages through digital innovation. The research in this paper is pragmatic and comprehensive.

(3) Previous studies have focused on issues related to digital innovation ecosystems via traditional econometric analyses but have ignored the objective spatial correlation between regions, which is not conducive to the high-quality evolution and capacity enhancement of digital innovation ecosystems. Using spatial convergence analysis, this paper not only explores the spatial convergence characteristics of the symbiotic level of digital innovation ecosystems in each region of China but also further examines the main factors affecting the spatial convergence of the symbiotic level of China's digital innovation ecosystems in terms of both economic structure and innovation application. This is an important addition to the application of methodology in the area of regional digital innovation.

5.3. Limitations and Prospects

Although the research in this paper has reached some conclusions of theoretical value and practical significance, there are still several limitations considering the complexity of the research problem. First, considering the availability of data, this study selected some publicly available secondary data to measure specific indicators on the basis of the relevant literature, which may have resulted in measurement bias. Future research can be extended and improved in the following two key directions. On the one hand, we can expand the data sources by adding microdata at the firm level and mesodata at the industry level to further clarify the synergistic paths and specific contributions of multiple types of players in regional digital innovation ecosystems. On the other hand, dynamic evolution is an important feature of regional digital innovation ecosystems, but the research period of this paper is limited to 2013–2022. Future research can continue to expand the time series data to better analyze the spatial and temporal evolution of the symbiotic level of regional digital innovation ecosystems.

Second, the research boundary of this paper focuses on the analysis of the symbiosis level of digital innovation ecosystems at the provincial level in the Chinese context. For more "regional" contexts, such as municipalities, economic development zones, and the Belt and Road Initiative, further investigations and explorations are needed. Therefore, future research could consider increasing the research boundary or expanding the scope of the study, especially the symbiosis of digital innovation ecosystems in developing and developed countries. This may result in significant differences in the symbiotic level of their digital innovation ecosystems due to differences in geographic background, innovation culture, national policies, and other conditions. Through comparative analyses across regions and countries, we may be able to obtain more robust and universal regional digital innovation ecosystem symbiosis theoretical systems and practical experience.

Author Contributions: Conceptualization, S.L., Z.L. and Y.W.; methodology, S.L. and Z.L.; formal analysis, Z.L. and Y.W.; writing—original draft preparation, S.L., Z.L. and Y.H.; writing—review and editing, S.L. and Z.L.; visualization, S.L., Z.L. and Y.H. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the National Natural Science Foundation of China (Grant No. 72204181; Grant No. 72402156; Grant No. 72002084), the Social Science Foundation of Jiangsu Province (Grant No. 22EYC005), the Ministry of Education of China's Project of Human-

ities and Social Sciences (Grant No. 21YJC630075), the Special Project on Financial and Economic Development of the Excellent Project of Social Science Application Research of Jiangsu Province (Grant No. 24SCB-44), the Key Project of Education Science Planning of Jiangsu Province (Grant No. B-b/2024/01/114; Grant No. B/2022/01/125), and the General Project of the 2024 Planning Project of the Commerce Statistical Society of China (Grant No. 2024STY23).

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The data presented in this study are available on request from the corresponding author.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Wang, H.H.; Zhou, J.; Guo, J.J.; Zhao, P.J.; Zhou, W.S. Regional Innovation Ecosystem Suitability, Dual Network and Innovation Performance: A Moderated Mediator Model. *Manag. Rev.* **2023**, *35*, 83–91.
2. Su, Y.; Chai, J.H.; Lu, S.C.; Lin, Z.Z. Evaluating green technology innovation capability in intelligent manufacturing enterprises: A Z-number-based model. *IEEE Trans. Eng. Manag.* **2024**, *71*, 5391–5409.
3. Xie, X.M.; Yu, J.H.; Tang, H.Y. A study of the impact mechanism of the population richness of innovation ecosystem on the innovation ecological effect. *Sci. Res. Manag.* **2022**, *43*, 9–21.
4. Endres, H.; Huesig, S.; Pesch, R. Digital innovation management for entrepreneurial ecosystems: Services and functionalities as drivers of innovation management software adoption. *Rev. Manag. Sci.* **2022**, *16*, 135–156. [CrossRef]
5. Abbate, T.; Codini, A.; Aquilani, B.; Vrontis, D. From knowledge ecosystems to capabilities ecosystems: When open innovation digital platforms lead to value co-creation. *J. Knowl. Econ.* **2022**, *13*, 290–304.
6. Xiao, W.X.; Fan, D.C. The "simultaneous development of quantity and quality": Research on the impact of the digital economy in enabling manufacturing innovation. *Systems* **2024**, *12*, 470. [CrossRef]
7. Qu, Y.Y. The Organizational Basis of Digital Innovation and China's Heterogeneity. *J. Manag. World* **2022**, *38*, 158–174.
8. Zhang, C.; Chen, K.H.; Mu, R.P. The digital innovation ecosystems: Theory building and a research agenda. *Sci. Res. Manag.* **2021**, *42*, 1–11.
9. Pierrakis, Y.; Saridakis, E. The role of venture capitalists in the regional innovation ecosystem: A comparison of networking patterns between private and publicly backed venture capital funds. *J. Technol. Transf.* **2019**, *44*, 850–873. [CrossRef]
10. Tian, Q.F.; Shen, W.K.; Li, Y. The evolution and impediments to regional digital innovation ecosystem from a symbiotic perspective. *Sci. Technol. Prog. Policy* **2024**, *41*, 1–12.
11. Ning, L.J.; Liu, J.T.; Xiao, Y.X.; Kong, D.J. Research on symbiosis model of digital innovation ecosystem. *Stud. Sci. Sci.* **2022**, *40*, 1481–1494.
12. Li, L.C.; Zheng, Y.J.; Peng, H.T. Research on the configuration path for the development of new quality productivity driven by digital innovation ecosystem. *Sci. Res. Manag.* **2024**, *45*, 1–10.
13. Lin, Y.; Liao, H. How can different types of technology-based startups be embedded in the digital innovation ecosystem? From the perspective of resource orchestration. *Study Explor.* **2023**, *4*, 98–107.
14. Liu, X.X. Evaluation of the symbiotic degree of China's digital innovation ecosystem: Dynamic evolution and influencing factors. *J. Tech. Econ. Manag.* **2024**, *3*, 49–54.
15. Gupta, R.; Mejia, C.; Kajikawa, Y. Business, innovation and digital ecosystems landscape survey and knowledge cross sharing. *Technol. Forecast. Soc. Change* **2019**, *147*, 100–109.
16. Brea, E. A framework for mapping actor roles and their innovation potential in digital ecosystems. *Technovation* **2023**, *125*, 102783.
17. Li, L.C.; Zeng, Y.J.; Xia, D. How does the digital innovation ecosystem enable green regional development? A dynamic QCA study in China. *Systems* **2024**, *12*, 551. [CrossRef]
18. Li, Y.H.; Fu, K.W.; Gong, X.H.; Xiang, Z.W.; Zhang, J.Y.; Liao, C.J. Research on value co-creation mechanism of platform enterprises in digital innovation ecosystem: A case study on Haier HOPE platform in China. *Front. Psychol.* **2022**, *13*, 1055932.
19. Ji, H.M.; Zou, H.; Liu, B.T. Research on dynamic optimization and coordination strategy of value co-creation in digital innovation ecosystems. *Sustainability* **2023**, *15*, 7616. [CrossRef]
20. Rong, Y.; Qiu, R.; Wang, L.Y.; Yu, L.Y.; Huang, Y.T. An integrated assessment framework for the evaluation of niche suitability of digital innovation ecosystem with interval-valued Fermatean fuzzy information. *Eng. Appl. Artif. Intell.* **2024**, *138*, 109326.
21. Pujadas, R.; Valderrama, E.; Venters, W. The value and structuring role of web APIs in digital innovation ecosystems: The case of the online travel ecosystem. *Res. Policy* **2024**, *53*, 104931. [CrossRef]

22. He, Y.X.; Song, J.H.; Ouyang, W.J.; Li, Q.H. Formation mechanism and implementation path of a digital agriculture innovation ecosystem. *Teh. Vjesn.-Tech. Gaz.* **2024**, *31*, 402–411.
23. Li, Y.; Wang, Y.T.; Wang, L.; Xie, J.C. Investigating the effects of stakeholder collaboration strategies on risk prevention performance in a digital innovation ecosystem. *Ind. Manag. Data Syst.* **2022**, *122*, 2045–2071. [CrossRef]
24. Li, M.; Zhu, J.H.; Dong, H. A study on the impact of watershed compensation policies on green technology innovation ecosystems. *Systems* **2025**, *13*, 44. [CrossRef]
25. Wolfert, S.; Verdouw, C.; van Wassenaer, L.; Dolfsma, W.; Klerkx, L. Digital innovation ecosystems in agri-food: Design principles and organizational framework. *Agric. Syst.* **2023**, *204*, 103558. [CrossRef]
26. Randhawa, K.; Vanhaverbeke, W.; Ritala, P. Legitimizing digital technologies in open innovation ecosystems: Overcoming adoption barriers in Healthcare. *Calif. Manag. Rev.* **2024**, *67*, 45–68. [CrossRef]
27. da Rosa, F.S.; Lunkes, R.J.; Schäfer, J.D.; Codesso, M.M. Digital innovation for food waste reduction in hotels: The complementary effect of digital capabilities and innovation ecosystem coopetition. *J. Sustain. Tour.* **2024**, 1–15. [CrossRef]
28. Wang, Z.Z.; Xu, S.Q.; Guan, Y.J. Impact of the innovation promotion strategy on digital technology diffusion in regional innovation ecosystems. *Technol. Anal. Strateg. Manag.* **2024**, 1–17. [CrossRef]
29. Sun, Y.W.; Zhou, Y.T. Specialized complementary assets and disruptive innovation: Digital capability and ecosystem embeddedness. *Manag. Decis.* **2024**, *62*, 3704–3730. [CrossRef]
30. Liu, J.T.; Ning, L.J.; Gao, Q.F. Research on the mechanism of digital innovation ecosystem embeddedness on the digital innovation performance of complementary enterprises: Evidence from China. *Kybernetes* 2024, *ahead of print*. [CrossRef]
31. Liu, Y.J.; Li, M.F. Analyzing the impact of digital innovation ecosystem on the intelligent development in high-end equipment manufacturing industry: A dynamic QCA analysis. *Bus. Process Manag. J.* 2024, *ahead of print*. [CrossRef]
32. Li, X.D.; Zhang, X.Y. Research on the influence of regional innovation ecosystem symbiosis on regional sci-tech innovation. *Stud. Sci. Sci.* **2019**, *37*, 909–918+939.
33. Chen, D.L.; Fu, M.; Wang, L. Study on the symbiosis evolution mechanism of the digital innovation ecosystem: Considering government regulation. *Kybernetes* **2025**, *54*, 3023–3039. [CrossRef]
34. Liu, J.T.; Ning, L.J.; Gao, Q.F. How can multi-agent collaboration achieve high digital innovation performance? A configuration study based on the perspective of digital innovation ecosystem. *J. Northeast. Univ. (Soc. Sci.)* **2024**, *26*, 52–64.
35. Suseno, Y.; Laurell, C.; Sick, N. Assessing value creation in digital innovation ecosystems: A social media analytics approach. *J. Strateg. Inf. Syst.* **2018**, *27*, 335–349.
36. Chae, B. A General framework for studying the evolution of the digital innovation ecosystem: The case of big data. *Int. J. Inf. Manag.* **2019**, *45*, 83–94. [CrossRef]
37. Wang, P. Connecting the parts with the whole: Toward an information ecology theory of digital innovation ecosystems. *MIS Q.* **2021**, *45*, 397–422. [CrossRef]
38. Buhe, C.L.; Chen, L. Digital innovation ecosystem: Concept, structure and operating mechanism. *Forum Sci. Technol. China* **2022**, *9*, 54–62.
39. Li, H.Y. The formation mechanism and implementation path of digital agriculture innovation ecosystem. *Issues Agric. Econ.* **2022**, *5*, 49–59.
40. Yang, W.; Lao, X.Y.; Zhou, Q.; Zhang, L. The governance niche configurations for the resilience of regional digital innovation ecosystem. *Stud. Sci. Sci.* **2022**, *40*, 534–544.
41. Zhao, T.Y.; Wang, H.Q.; Li, Y.; Deng, Q.Y. Formation mechanism of the comprehensive advantage of emerging industry innovation ecosystem: A case study of new energy vehicle industry. *Stud. Sci. Sci.* **2023**, *41*, 2267–2278.
42. Ge, X.C.; Ji, L. Analysis on formation and evolution of the innovation ecosystem of digital economy industry. *Reform. Econ. Syst.* **2023**, *1*, 125–134.
43. Jing, L.L.; Huang, H.L. The stimulation and impact of digital innovation ecosystem on regional innovation capability in two dimensions of time and space: A dynamic QCA analysis based on provincial panel data. *Sci. Technol. Prog. Policy* **2024**, *41*, 13–23.
44. Li, X.D.; Rao, M.X. Research on the development path of the regional digital innovation ecosystem: Configuration analysis based on fsQCA. *J. Ind. Eng. Eng. Manag.* **2023**, *37*, 20–31.
45. Sun, Y.L.; Zhu, R.J.; Song, J. Research on the evolution and governance of digital innovation ecosystem. *Stud. Sci. Sci.* **2023**, *41*, 325–334.
46. Wei, J.; Zhao, Y.H. Governance mechanism of digital innovation ecosystem. *Stud. Sci. Sci.* **2021**, *39*, 965–969.
47. He, Z.C.; Li, Y.J.; Bai, M.J.; Pan, W.H. The construction of a logarithmic intelligent innovation ecosystem in core enterprises: A case study of Huawei automotive. *Theory Pract. Financ. Econ.* **2024**, *45*, 100–108.
48. Ma, R.L.; Liu, H.; Li, Z.P.; Ma, Y.F.; Fu, S.L. Promoting sustainable development: Revisiting digital economy agglomeration and inclusive green growth through two-tier stochastic frontier model. *J. Environ. Manag.* **2024**, *355*, 120491. [CrossRef]
49. Beltagui, A.; Rosli, A.; Candi, M. Exaptation in a digital innovation ecosystem: The disruptive impacts of 3D printing. *Res. Policy* **2020**, *49*, 103833. [CrossRef]

50. Granstrand, O.; Holgersson, M. Innovation ecosystems: A conceptual review and a new definition. *Technovation* **2020**, *21*, 90–91.
51. Hu, Y.L.; Bai, S.Z. Research on the multi-agent cooperative innovation supernetwork model and governance path of digital ecosystem. *Forum Sci. Technol. China* **2024**, *10*, 53–62.
52. Chen, H.M.; Cai, S.L. Evaluation and spatial-temporal evolution of regional digital innovation ecosystem resilience. *Stat. Decis.* **2023**, *39*, 51–55.
53. Ceccagnoli, M.; Forman, C.; Huang, P. Cocreation of value in a platform ecosystem! The case of enterprise software. *MIS Q.* **2012**, *36*, 263.
54. Yang, X.; Li, X.P.; Zhou, D.C. Study on the difference and convergence of carbon productivity in Chinese manufacturing. *J. Quant. Technol. Econ.* **2015**, *32*, 3–20.
55. Su, Y.; Yan, Y.H. The influence of the two-tier network of a regional innovation system on knowledge emergence. *J. Knowl. Manag.* **2023**, *27*, 2526–2547. [CrossRef]
56. Fan, D.C.; Wu, X.L. Spatial correlation network analysis of industrial green technology innovation efficiency in China. *Systems* **2023**, *11*, 240. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.